

Probabilistic measurement of neutron spectra with nFoils

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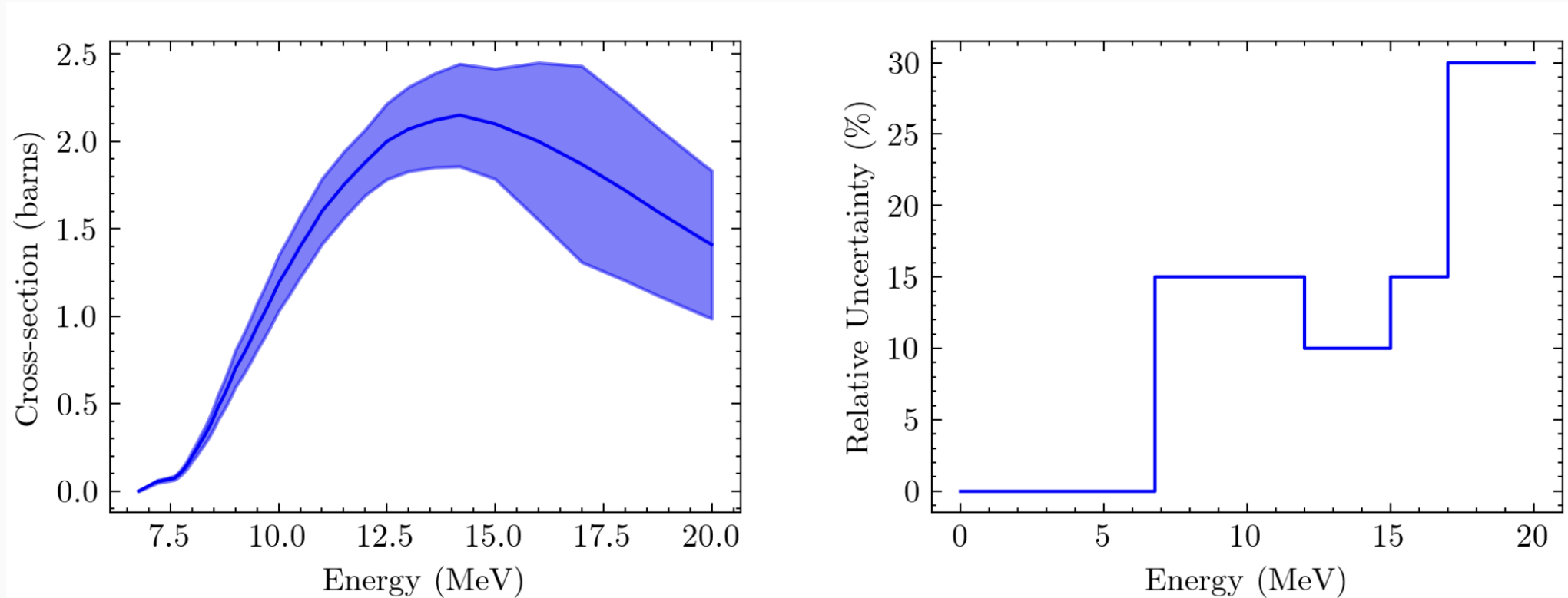
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Nuclear data

Nuclear data underpins all neutronics simulations. Rubbish data → rubbish simulation results



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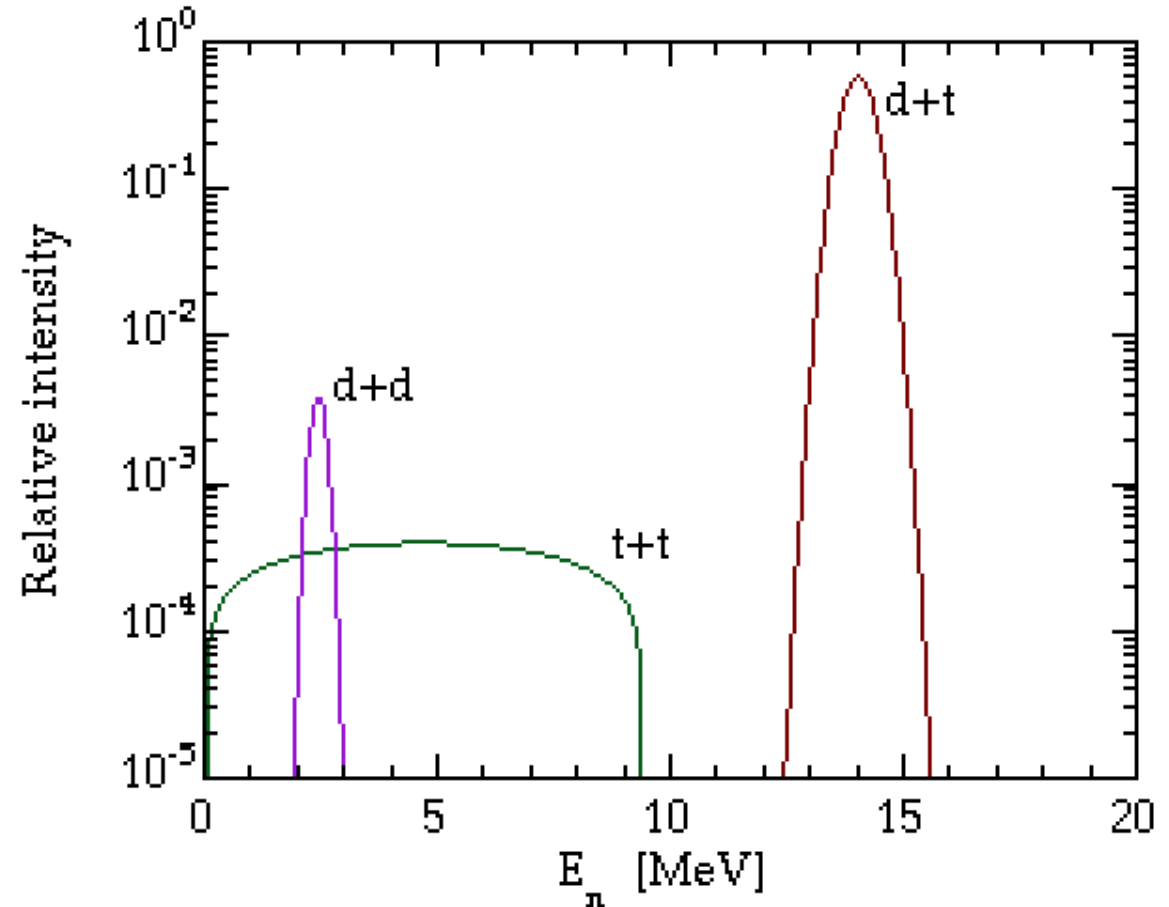
Uncertainty in the $\text{Pb}(n,2n)$ cross-section in the ENDF/B-V library.
Tritium breeding blanket predictions are sensitive to this cross-section
(E. Miralles-Dolz 2024 *University of Liverpool* (Thesis))

Neutron sources

Fast neutron sources are useful for investigating activation or transport **nuclear data**

Either a broad or peaked (i.e. DT) neutron spectrum

More precisely measured neutron spectrum → more precise experiments → more confidence in conclusions



Contributions to the fusion reactor neutron energy spectrum
(K. Leung *et al* 2004 LBNL (Report))



Measuring neutron spectra

A neutron spectrum can be '**unfolded**' by measuring how it activates isotopes in a set of thin foils

These measurements are called reaction rates

If one also knows the reaction response functions and some prior knowledge, the spectrum can be estimated

N_i = reaction rate in reaction channel i

R_{ij} = response matrix element for reaction channel i and energy channel j

$$N_i = \sum_{j=1}^n R_{ij} \Phi_j$$

m = total number of reaction channels
 n = total number of energy channels

Φ_j = neutron flux in energy channel j



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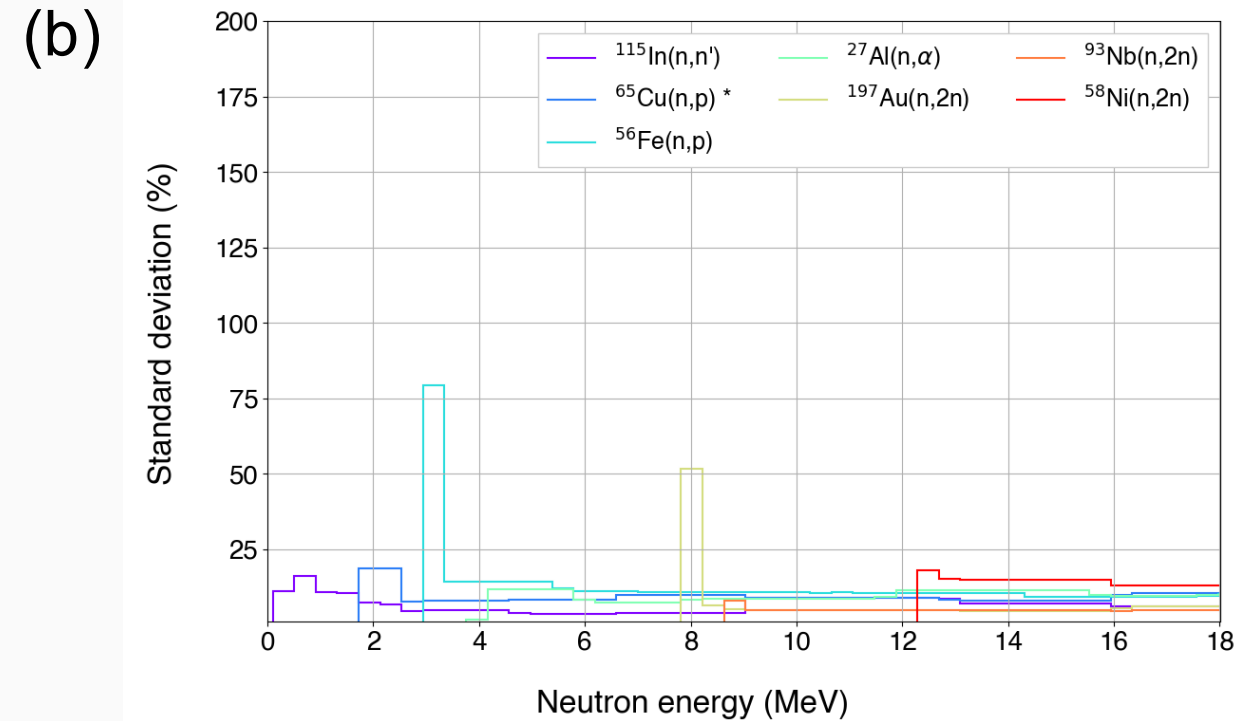
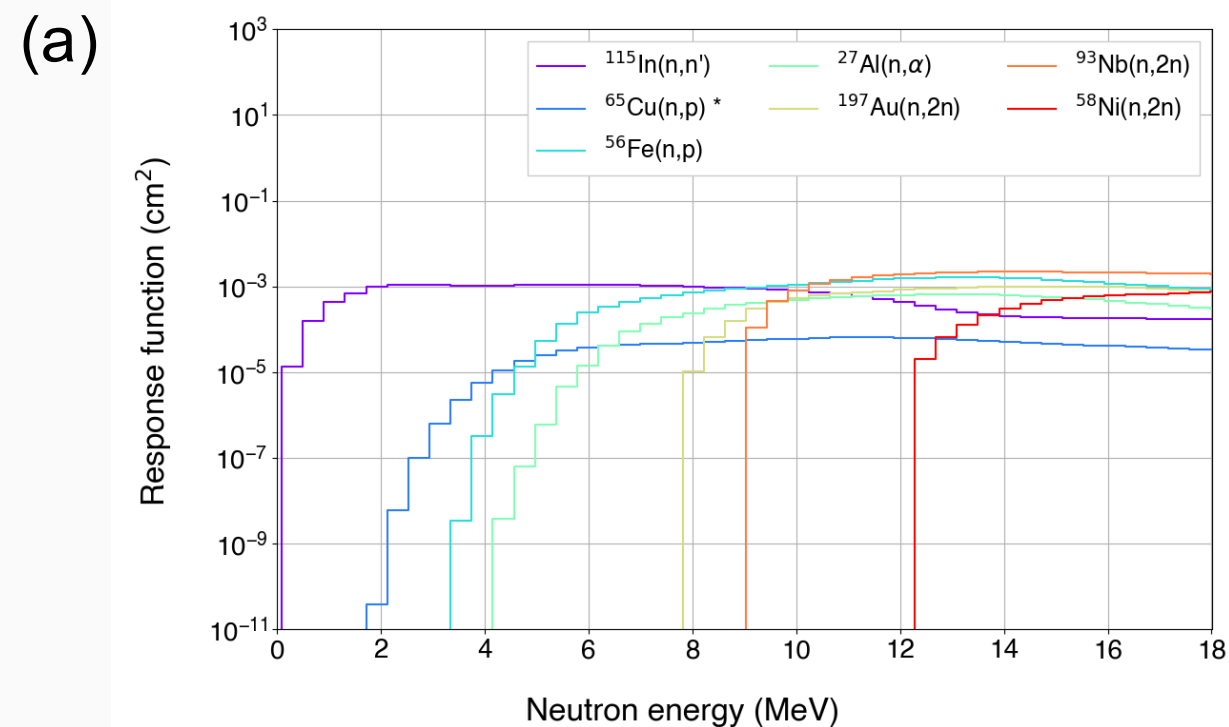
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Discretised unfolding Fredholm equations for a single reaction channel i
 (R. Worrall *et al* 2020 *Fus. Eng. Design* **161** 112038)

Uncertainties

Dosimetry foil unfolding has no unique solution → uncertainties are very important

Uncertainty quantification is **not well done** for complex spectra in fine group structures



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TENDL-2021 reaction (a) response functions and (b) uncertainties used in our validation experiment (* indicates ENDF/B-VIII)

Bayesian unfolding

1. Approximate the prior spectrum with a set of adjustable parameters
2. Construct a likelihood from the measurements
3. Create a response matrix distribution to sample over
4. Use MCMC to find the posterior distributions for the spectrum parameters

Likelihood distribution for reaction rates $\vec{N}$, given the prior parameters $\vec{\alpha}$ and the known response matrix $\vec{R}$

Prior distribution for the spectrum parameters $\vec{\alpha}$, given some limits

$$P(\vec{\alpha} \mid \vec{N}, \vec{R}) \propto P(\vec{N} \mid \vec{\alpha}, \vec{R}) P(\vec{\alpha})$$

Posterior distribution for the parameters $\vec{\alpha}$, given the reaction rates $\vec{N}$ and response matrix $\vec{R}$



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Bayes theorem for parametric inference of a neutron spectrum
(M. Reginatto 2010 *Rad. Meas.* **45**(10) 1323)

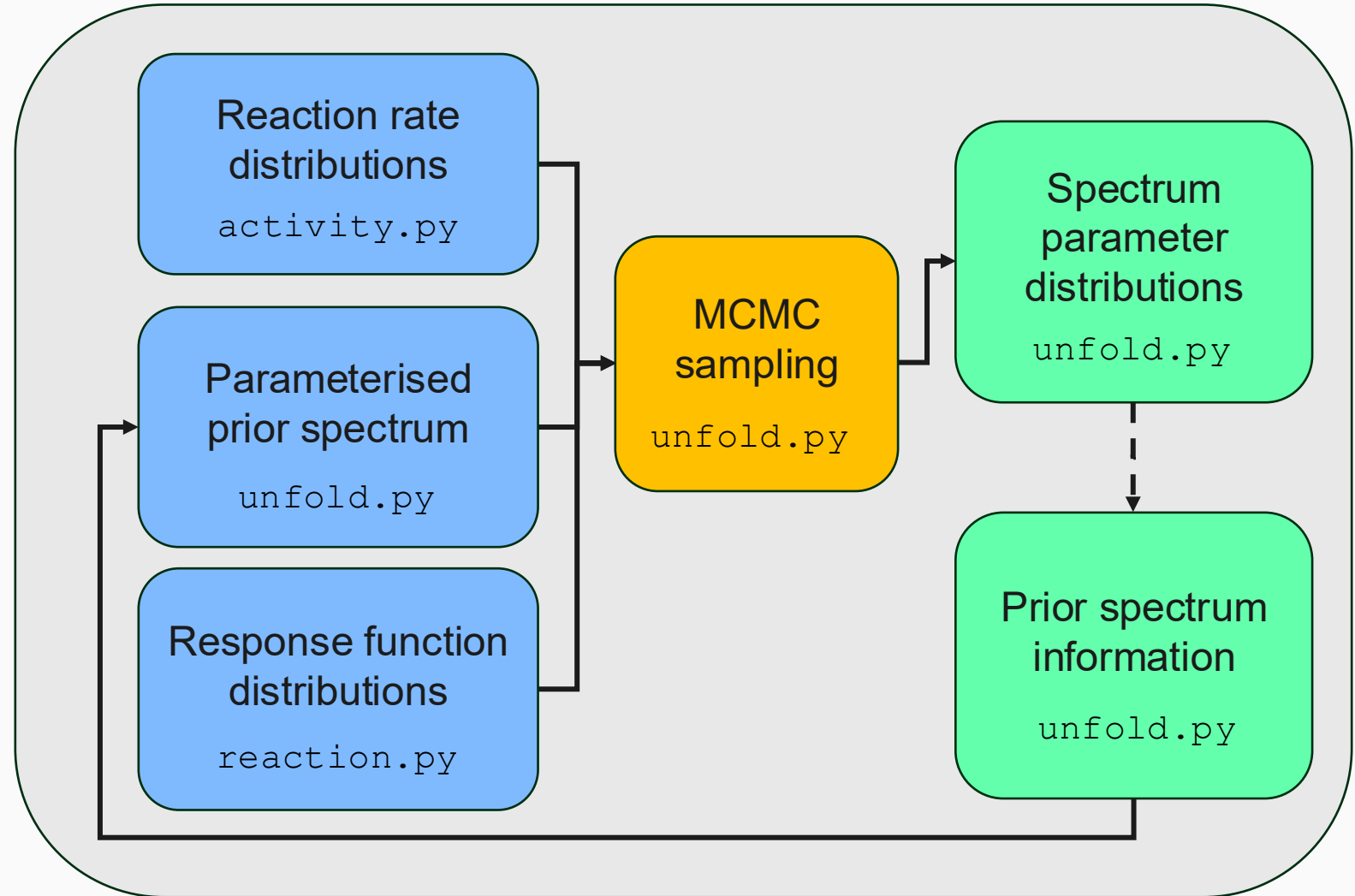
The nFoils package

An open source Python toolkit was developed for implementation

- Transparent methods
- Reproducible analysis
- Available to anyone

See more at

github.com/louisbutt338/nFoils



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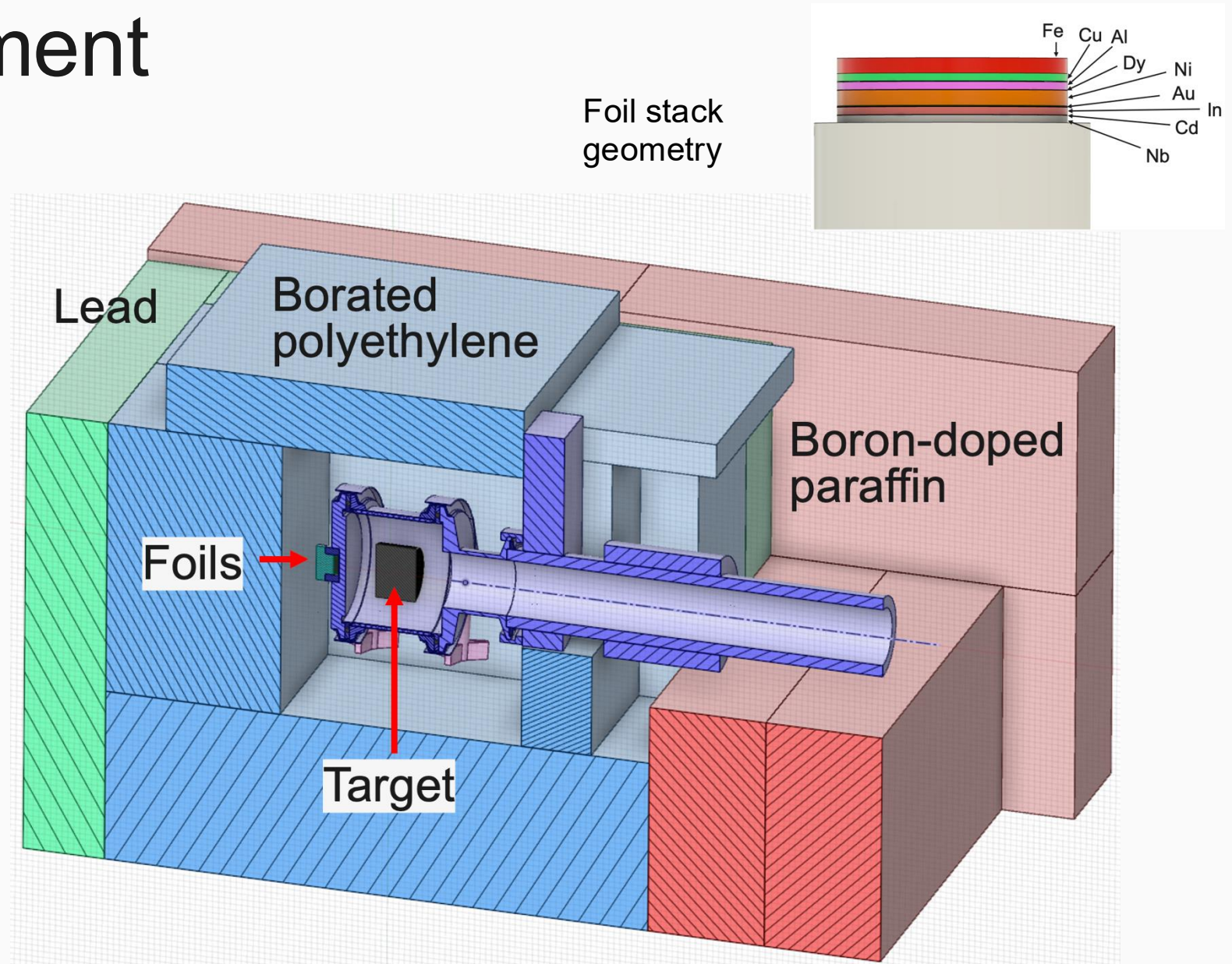
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Validation experiment

Neutrons from 16 MeV protons (p) on a lithium (Li) target were measured at our cyclotron

Gamma spectroscopy was used to measure foil activities

A CAD-to-MCNP model was developed to model the spectrum



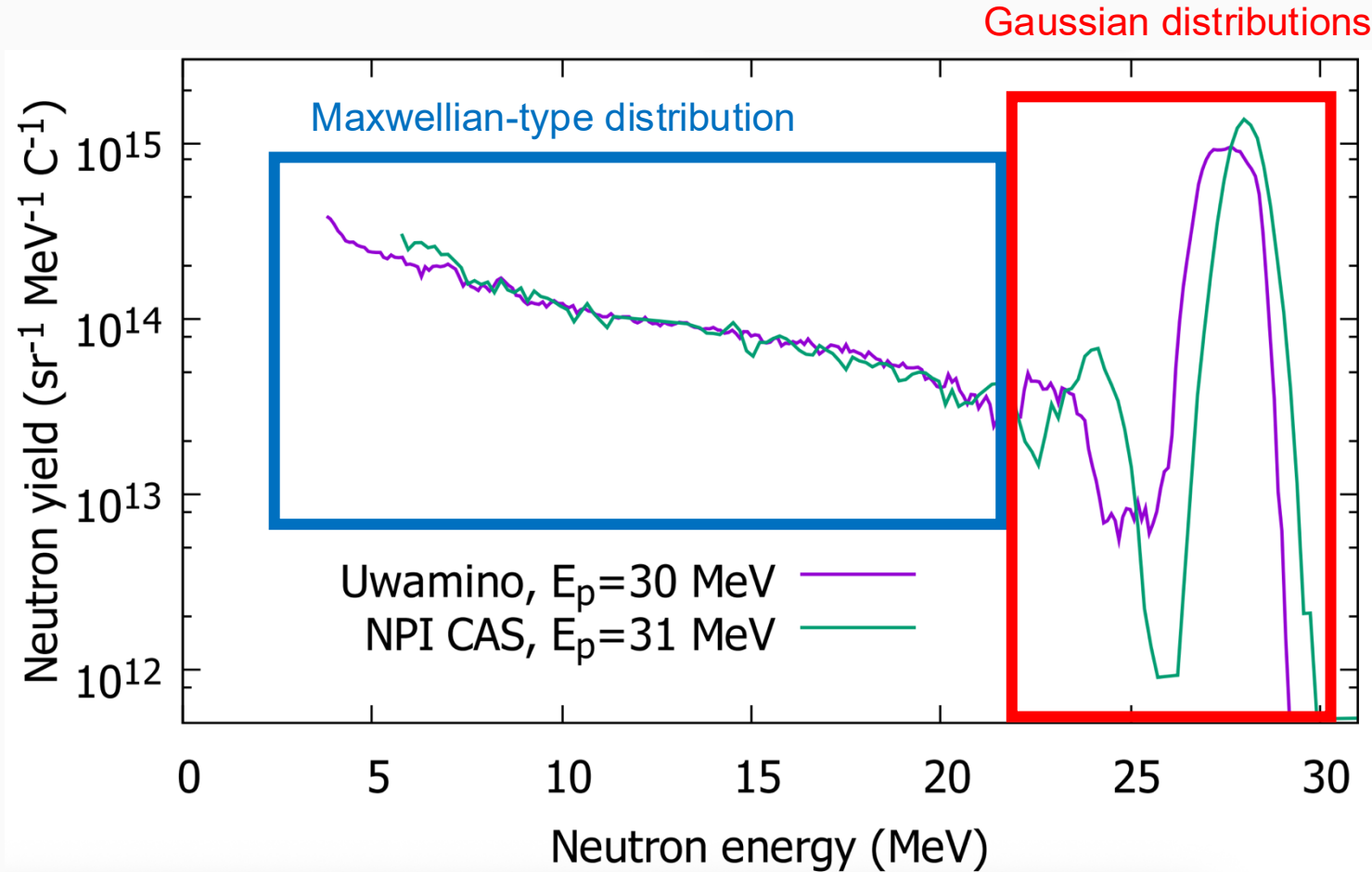
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Cross section of the model of the experiment

Spectrum prior



We must describe the spectrum as a set of distributions with parameters $\vec{\alpha}$

The p-Li neutron spectrum is:

- **Discrete peaks** from ${}^7\text{Li}(p,n){}^7\text{Be}$ direct reactions
- **Continuum** from ${}^7\text{Li}(p,n){}^3\text{He}, {}^4\text{He}$ compound nucleus evaporation



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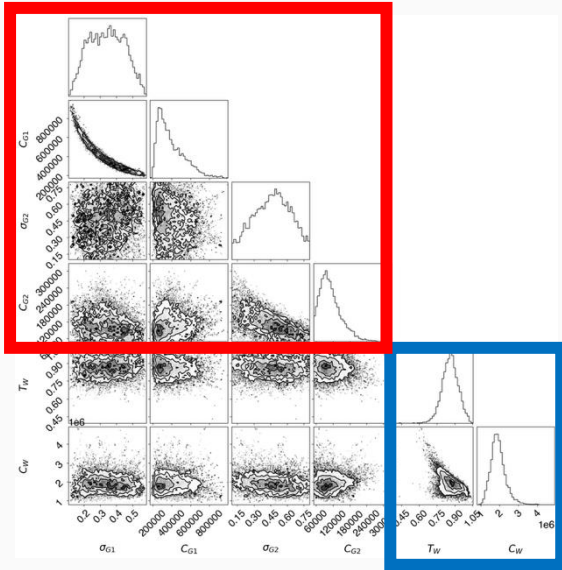


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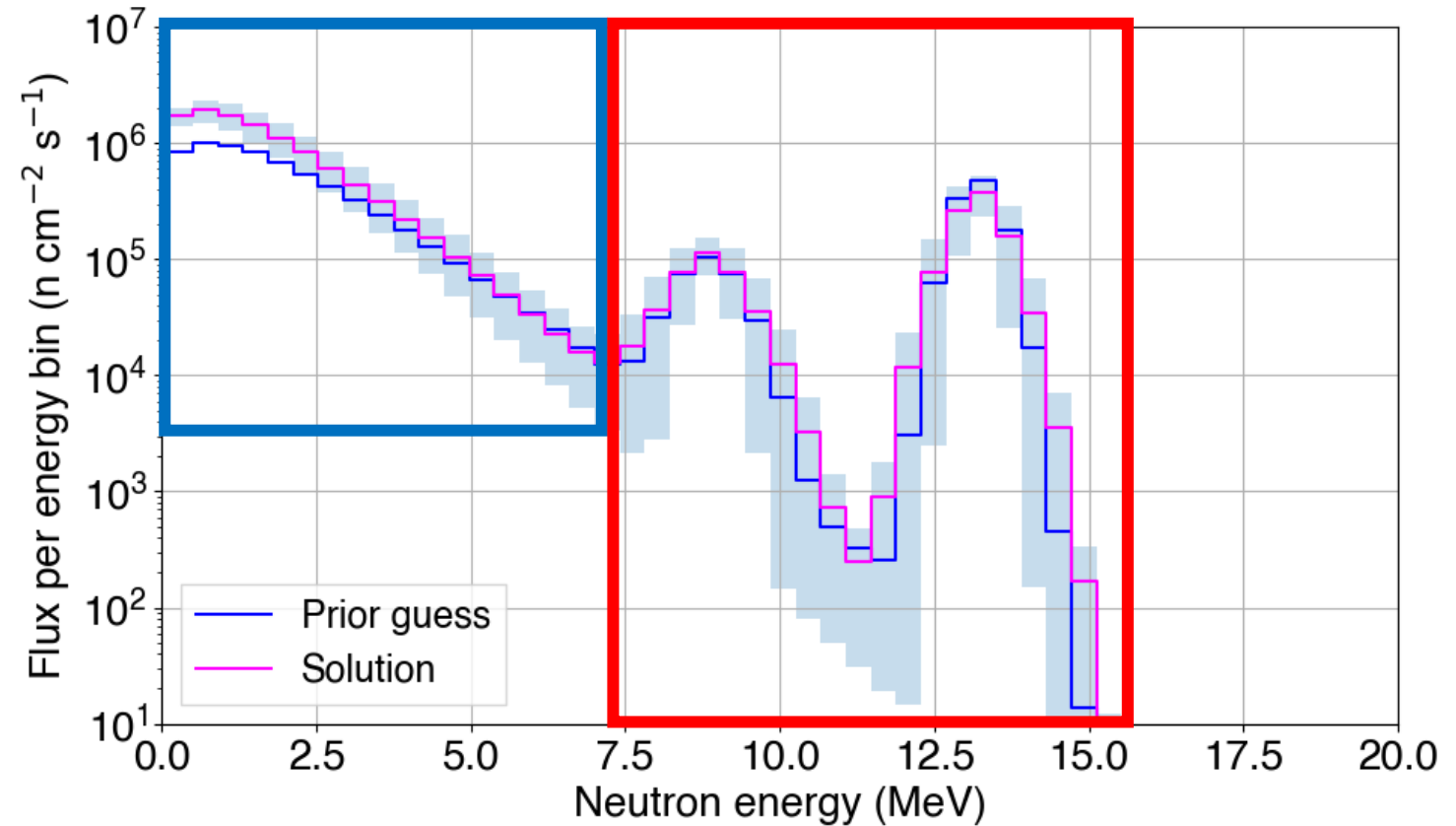
Measured neutron spectra from p-Li
(M. Majerle *et al* 2020 *EPJ Web of Conferences* **239**
20010; Y. Uwamino *et al* 1997 *NIM-A* **389(3)** 463-473)

Measured spectrum

Here two fixed **Gaussian** peaks are used for two discrete peaks, and one **Weisskopf** distribution for the continuum



Posterior distributions for the spectrum parameters



Spectrum unfolded with our algorithm in a linearly spaced 50-bin group structure. Accuracy is better than the MCNP model, but uncertainty is very high



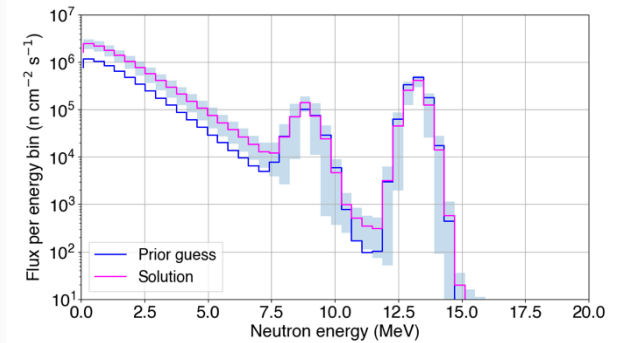
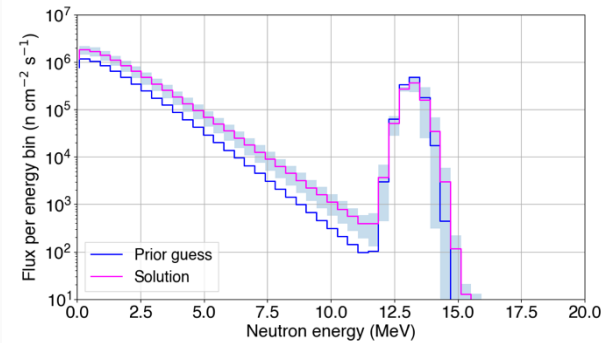
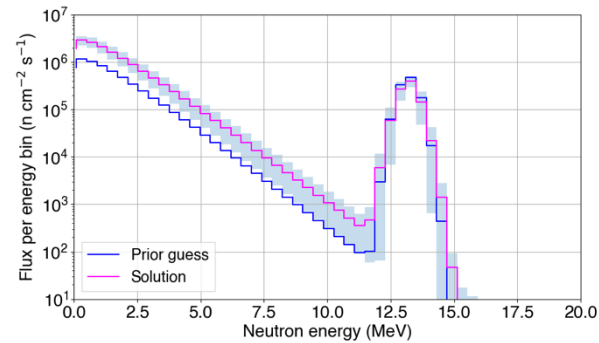
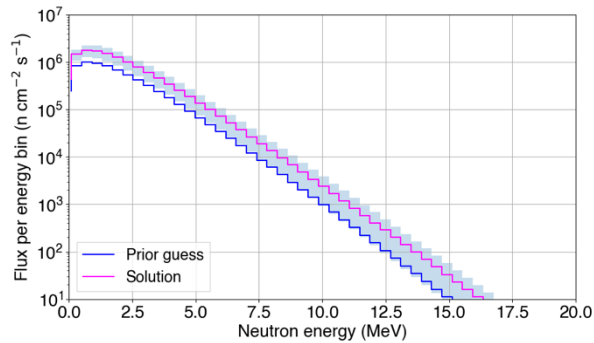
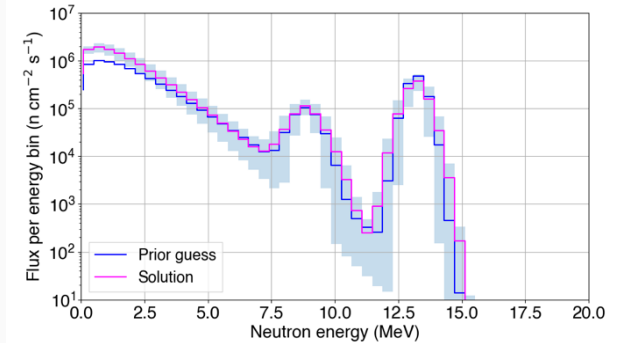
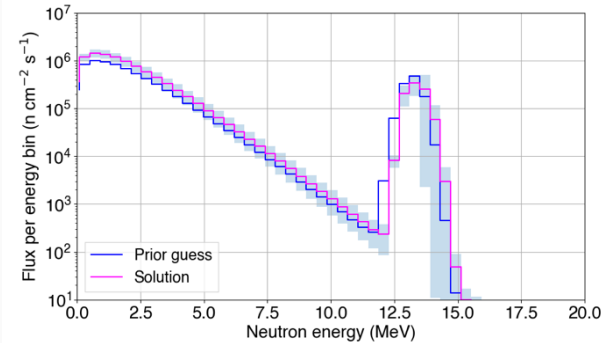
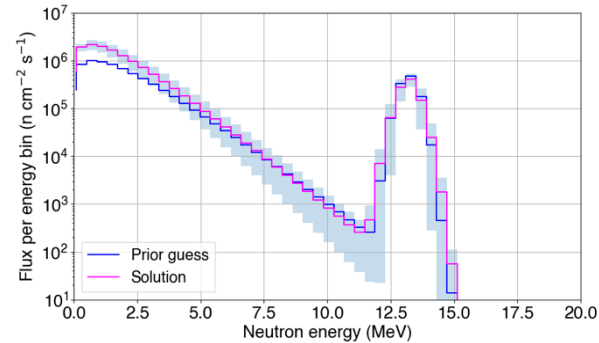
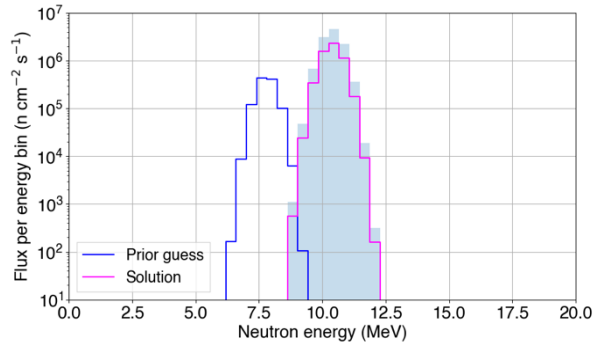
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Sensitivity analysis

How sensitive is the measurement to variable user inputs?



Too simple

Most accurate

Most precise

Too complex



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Sensitivity of the unfolded spectrum to the choice of prior. The priors are combinations of fixed/free Gaussians for the discrete peaks, and Weisskopfs/Maxwellians for the continua, plus a limit for unphysical neutron energies

Key takeaways

“[...] recommending that scientists use Bayes’ theorem is like giving the neighborhood kids the key to your F-16.”

(A. Gelman *et al* 2008 *Bay. Analysis.* **3(3)** 445)

Advantages +

Spectrum uncertainty is fully captured

Good prior knowledge can be utilised

Input data, group structure, prior model are all adjustable

Limitations -

Spectrum uncertainty is fully captured!

An airtight physics-informed prior is **critical**

Measurement is limited by:

- Choice of prior/spectrum complexity
- Number of measurements
- Group structure



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Call to arms

Software is available now at github.com/louisbutt338/nFoils

More developers welcome to take the package in fun new directions, or test the methods further

Papers describing the methods out later this year

Come find me or email me at LJB841@bham.ac.uk if you want to know more!

Acknowledgements:

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